

TABLE OF CONTENTS

1. OBJECTIVE
2. TOOLS AND COMPONENTS REQUIRED
3. CIRCUIT DIAGRAM
4. PROGRAM CODE
5. WORKING
6. OUTPUT/RESULT 7.LEARNING OUTCOMES
7. PROBLEMS FACED AND THEIR SOLUTIONS
8. REFERENCES

LIST OF FIGURE

1.1 ARDUINO UNO BOARD 1.2 ARDUINO IDE INTERFACE 1.3 CIRCUIT DIAGRAM OF ARDUINO SETUP 2.1 SEVEN SEGEMENT LED INTERFACING CIRCUIT 2.2 COUNTER OUTPUT ON SEVEN SEGMENT DISPLAY 3.1 LED MATRIX INTERFACING CIRCUIT

LIST OF TABLE

1.1 ARDUINO UNO SPECIFICATIONS 1.2 COMPONENTS REQUIRED FOR EXPERIMENT 0 2.1 PIN CONNECTIONS OF SEVEN SEGMENT DISPLAY 2.2 LED MATRIX PIN CONFIGURATION 3.1 LED MATRIX PATTERN DATA 3.2 OBSERVATION AND RESULT TABLE

EXPERIMENT 0: ARDUINO BASIC PROGRAM

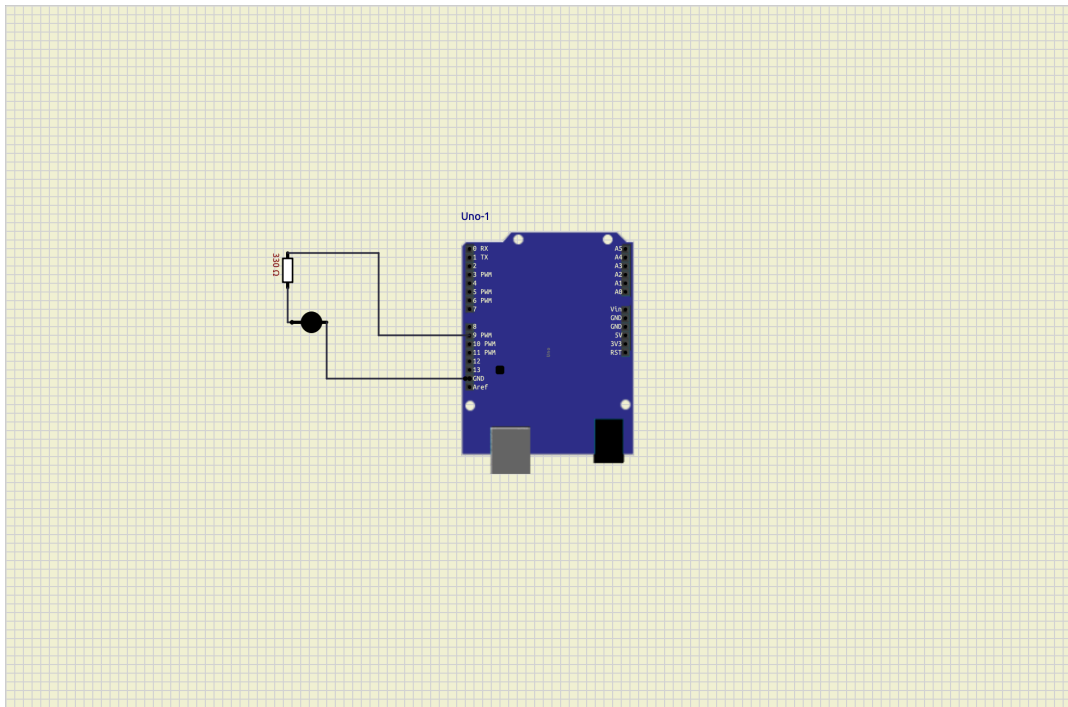
OBJECTIVE:

- learn the components of an arduino uno circuit .
- understand how to connect basic electronic components.
- write and upload a simple arduino program .
- verify the circuit through simulation .

TOOLS REQUIRED

1. Arduino uno/esp32
2. LED
3. Arduino IDE

CIRCUIT DIAGRAM



ARDUINO CODE

```
#include <const int ledPin = 9; // The digital pin connected to the LED

void setup() {
  // Initialize digital pin 13 as an output.
  pinMode(ledPin, OUTPUT);
}

void loop() {
  digitalWrite(ledPin, HIGH); // Turn the LED on
  delay(1000);                // Wait for 1 second (1000 milliseconds)
  digitalWrite(ledPin, LOW);  // Turn the LED off
  delay(1000);                // Wait for 1 second
}
```

WORKING

The program turns the LED ON for one second and OFF for one second repeatedly creating a blinking effect.

LEARNING OUTCOME

Learned how to connect an LED to Arduino Uno, write a basic Arduino program, and upload code to the board.

PROBLEMS FACED AND SOLUTIONS

1. Problem: LED did not blink. 2. Solution: checked wiring connections and upload the code again.

FUTURE SCOPE

This experiment can be extended to control multiple LEDs and other electronic components.

REFERENCES

1. Arduino IDE documentation
2. Arduino Uno datasheet

EXPERIMENT 01 :

INTERFACING WITH SEVEN SEGMENT LED TO MAKE A COUNTER USING A PUSH BUTTON AS INPUT

OBJECTIVE

To interface a seven segment LED display with Arduino and use a push button to increment and display a counter value.

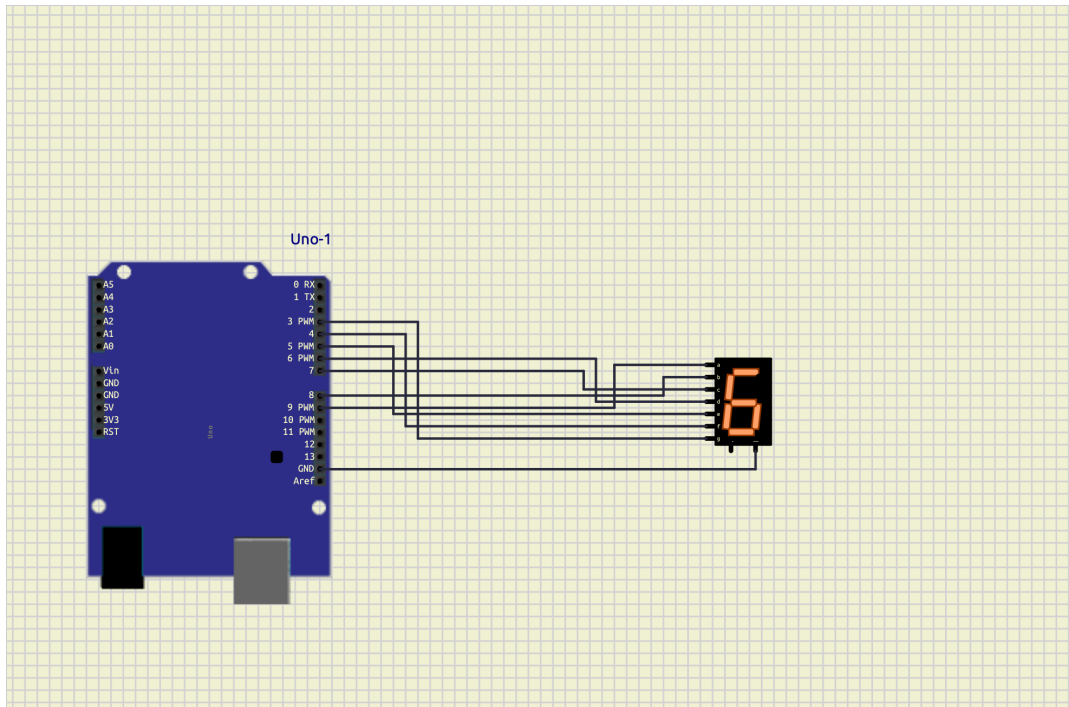
TOOLS REQUIRED

1. Arduino Uno
2. seven segment LED display (common cathode)
3. push button
4. Arduino IDE

THEORY

A seven segment display consists of seven LEDs arranged in the shape of the number 8. By turning ON and OFF specific segments, digits from 0 to 9 can be displayed. A push button is used as an input device. Each time the button is pressed, the counter value increases by one and the updated number is displayed on the seven-segment display.

CIRCUIT DIAGRAM



ARDUINO CODE

```

## include <sevseg.h>
void setup (){
    //put your setup code here ,to run once :
    byte numdigits =1;
    byte digit_pins []={};
    byte seg_pins []={9,8,7,6,5,4,3,2};
    byte dis_type =COMMON_CATHODE;
    bool res_on_segs=true;
    s_seg.begin (dis_type,numdigit_pins ,seg_pins ,res_on_segs
    s_seg.setbrightness(90);
}
void loop(){
    //put your main code here to run repeated:
    for (int i=0 ;i<10; i++)
    {
        s_seg.setnumber(i);
        s_seg.refreshdisplay ();
        delay(1000);
    }
}

```

PROCEDURE

1. Connect the seven -segment display to arduino digital pins .
2. connect the push button to the input pin.
3. upload the programto arduino uno .
4. press the push button repeatedly.
5. observe the display number (1 to 9)

OBSERVATIONS

1. The display initially show 0.
2. Each button press increments the counter.
3. after reaching 9,the counter rests to 0.

LEARNING OUTCOME

1. Learned how to interface a seven -segment display with arduino .
2. Understood digital input using a push button .
3. Implemented a simple counter system.

PROBLEM FACED AND SOULTIONS

PROBLEM: Incorrect digits displayedon the seven-segment display . SOLUTION:
Checked segment pin connections and corrected the writing according to the

circuit diagram .

FUTURE SCOPE

1. Two digital and four- digit counters.
2. visitor counting systems.
3. digital scoreboards.
4. electronic voting and counting applications .

REFERENCES

1.Arduino official Documentation 2. seven segment display datasheet 3 .arduino IDEuser guide

EXPERIMENT 03 DISPLAY DIFFERENT PATTERNS ON LED MATRIX

OBJECTIVE

To interface an LED matrix arduino and display different patterns ,symbols and designs on the matrix .

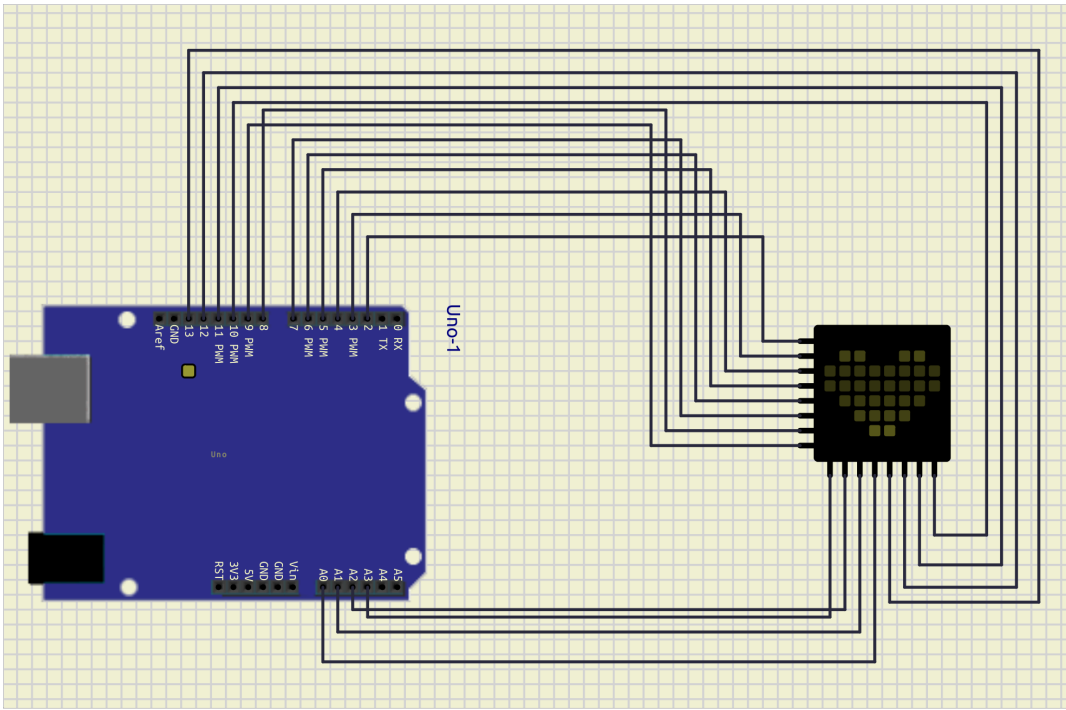
TOOLS REQUIRED

1. arduino uno
2. 8x8 LED matrix
3. max 7219 led matrix driver module (if used)

THEORY

An LED matrix is a grid of LED arranged in rows and columns . by controlling the LEDs individually ,different patterns ,symbols ,charavters, and animations can be displayed the arduino sends data to the LEDmatrix to illuminate specific LEDs and create the desired pattern .

CIRCUIT DIAGRAM



ARDUINO CODE

```

int pins[] = {2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13}; // Add

void setup() {
  for (int i = 0; i < 12; i++) {
    pinMode(pins[i], OUTPUT); // Fixed capital M and capitalize
  }
}

void loop() {
  for (int i = 0; i < 12; i++) {
    digitalWrite(pins[i], HIGH); // Fixed capital W
  }
  delay(1000); // Optional: adds a pause while they are ON

  for (int i = 0; i < 12; i++) {
    digitalWrite(pins[i], LOW); // Fixed capital W and capitali
  }
  delay(1000); // Optional: adds a pause while they are OFF
}

```

PROCEDURE

1. Connect the LED matrix to the arduino .
2. upload the program using arduino IDE.
3. Run the circuit .
4. observe the displayed pattern on the LED matrix .
5. modify the pattern data to display different shapes and symbols

OBSERVATIONS

1. The led matrix successfully displayed the programmed pattern
2. different patterns can be created by changing the binary values
3. the matrix can display symbols ,letters, numbers, and simple

LEARNING OUTCOME

1. learned how to interface an LED matrix with arduino
2. understood row and column addressing in LED matrices.
3. Gained experience in creating visual patterns using program

PROBLEMS

pattern was not displayed correctly .

SOLUTION:

Checked wiring connections and verified the binary pattern data

FUTURE SCOPE

1. Display scrolling text .
2. create animations and games
3. Design digital notice boards

REFERENCES

1. Arduino official documentation
2. max 7219 led matrix datasheet
3. Arduino LED matrix library documentation